

BactiMate

Your mate for bacteria microscopy

From raw data to publication — one workflow.

Why BactiMate

One application. Every step of your pipeline.

1 **TIFF Conversion**

ND2, LIF, CZI → TIFF. Metadata, drift & photobleaching.

2 **Omnipose Segment**

Built-in & custom models. Preview, batch, train your own.

3 **U-NET Segment**

U-Net or Building U-NET. Preview and batch prediction.

4 **Analyze Masks**

Morphology filters, Excel, fluorescence & foci detection.

5 **Tracking**

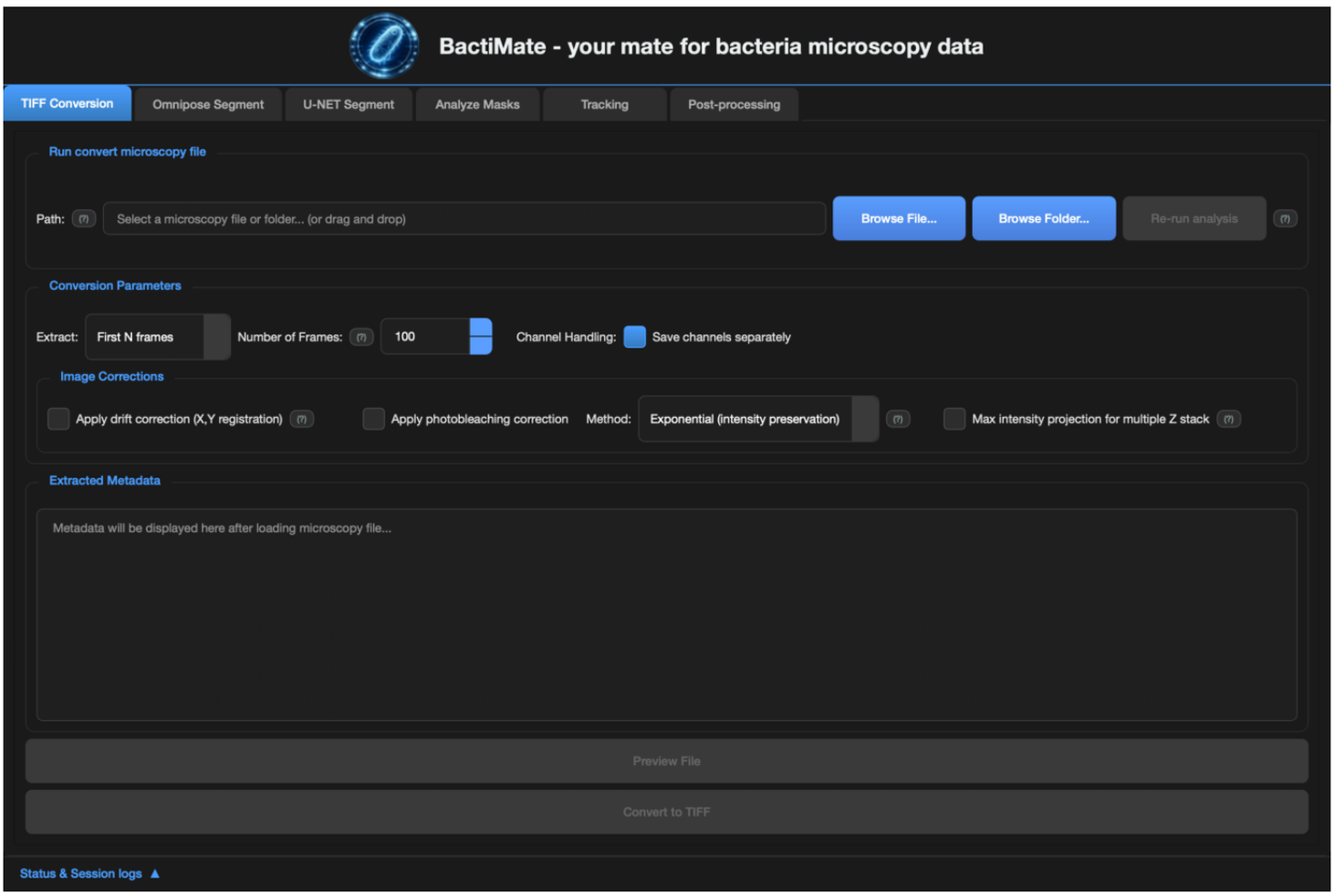
Trackastra lineage. Persistent IDs. Link to analysis.

6 **Post-processing**

Scale bars, montages, growth-curve videos.

TIFF Conversion

Convert once. Analyze everywhere.

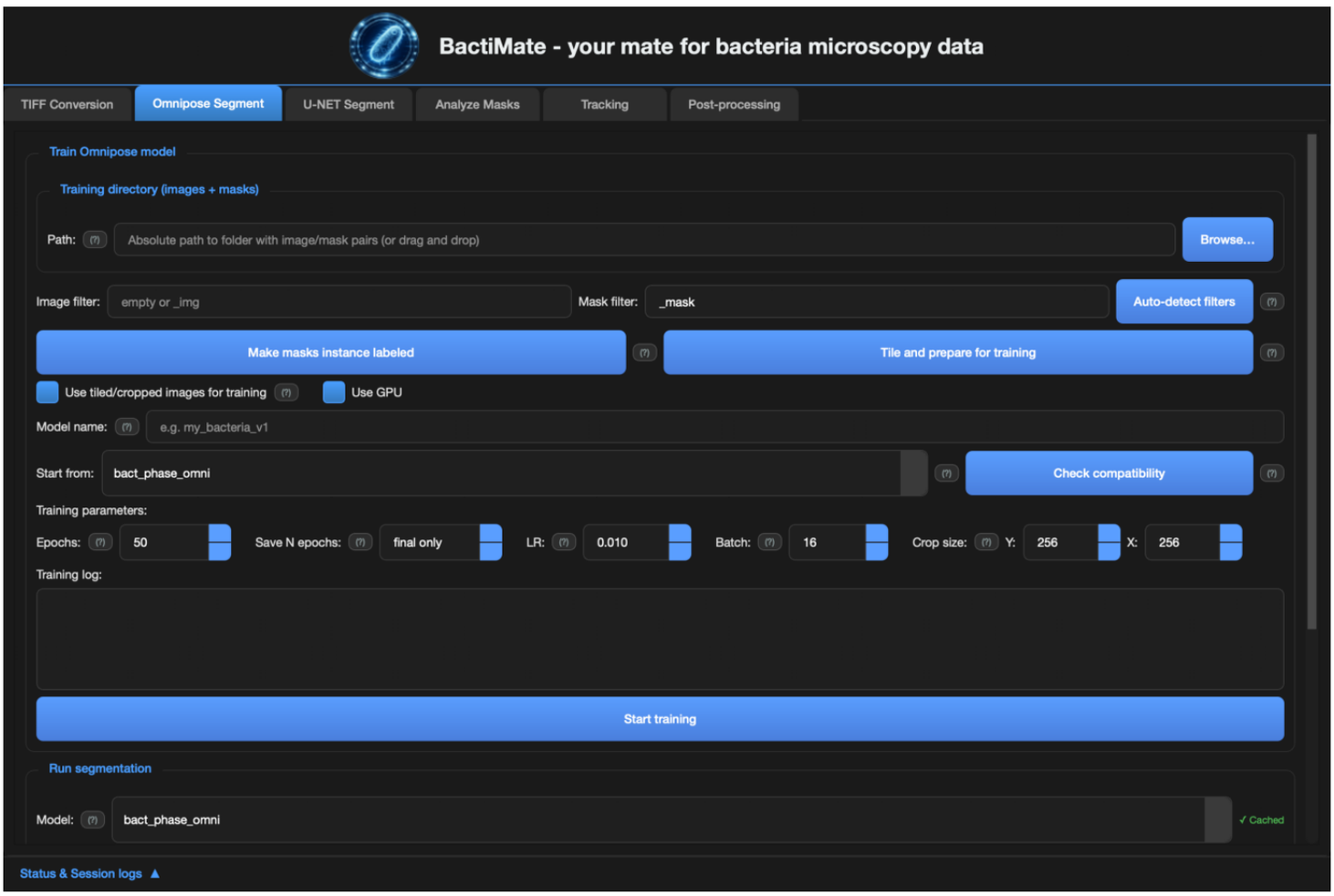


Turn ND2, LIF, and CZI files into TIFF with one workflow. Your pipeline starts here.

- Set frame count and channel handling (separate or combined)
- Optional drift correction (X,Y registration) and photobleaching correction
- Preview before convert; batch conversion for folders
- Mark channels as FI or Foci for Analyze Masks

Omnipose Segment

State-of-the-art cell segmentation.

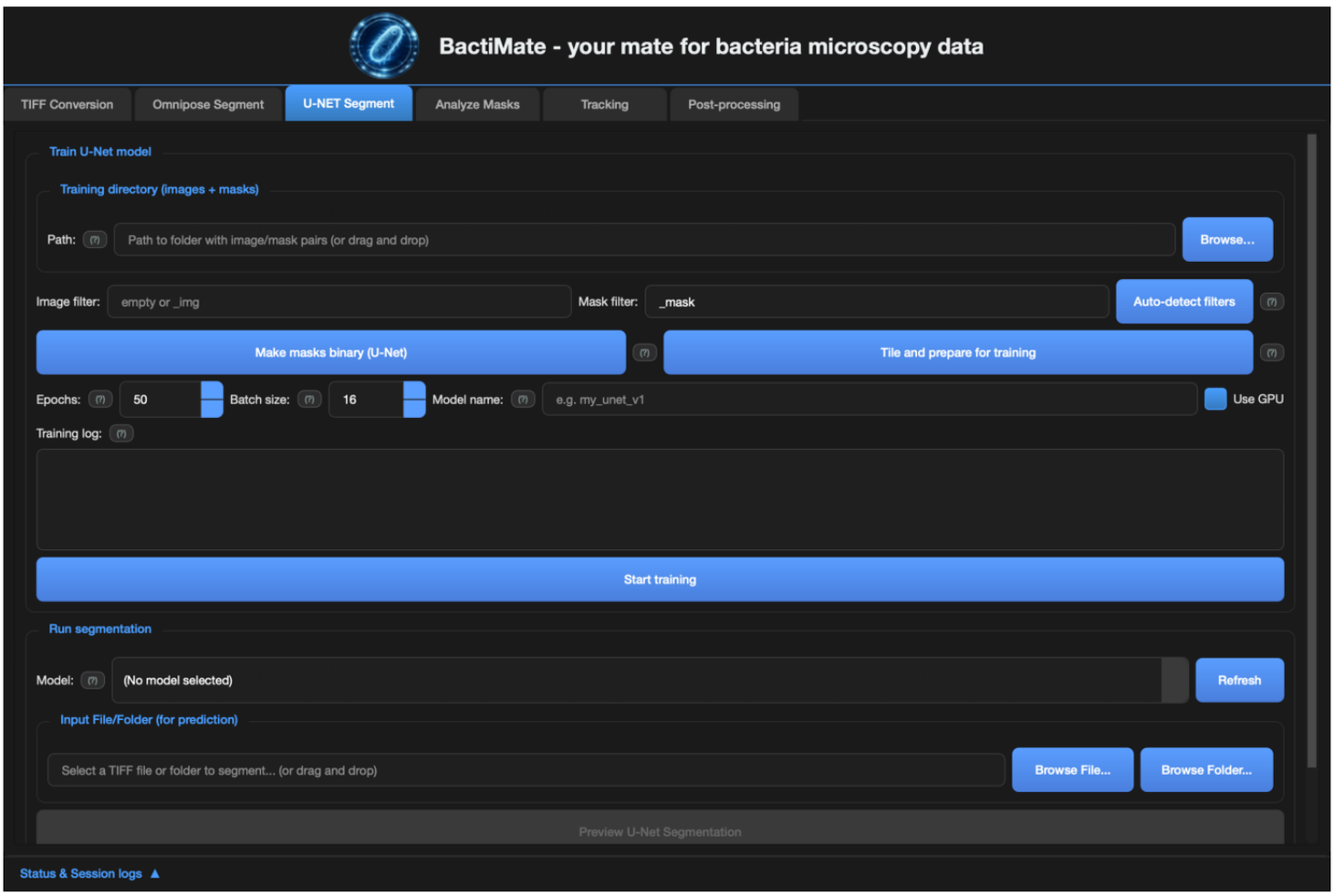


Segment bacteria with built-in or your own trained models. Live preview, batch, or train.

- Choose built-in model (e.g. bact_phase_omni) or your trained .pth model
- Diameter (0=auto), flow & mask threshold; GPU, DBSCAN, overlay
- Preview with overlay and frame navigation; Process file or folder
- Train custom model from image/mask pairs; transfer learning supported

U-NET Segment

Your model. Your workflow.

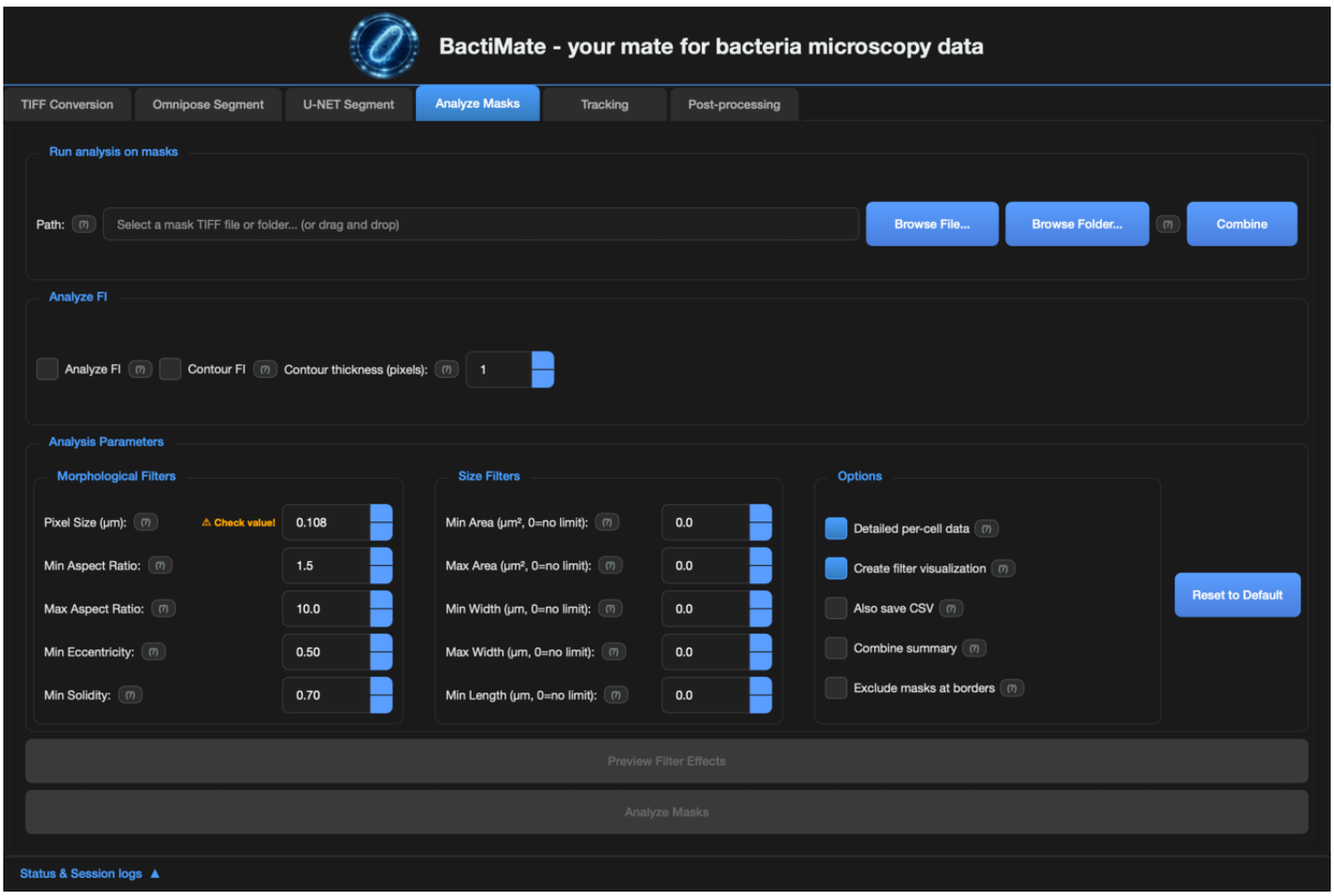


Prefer U-Net? Train in BactiMate or use Building U-NET. Preview and batch.

- Model dropdown: trained .keras from BactiMate or train new
- Training: image/mask pairs, filters, tile & prepare; optional Building U-NET path
- Preview U-NET Segmentation; Process file or folder
- Output to _segmentation_unet/ or next to input

Analyze Masks

From masks to metrics in one click.

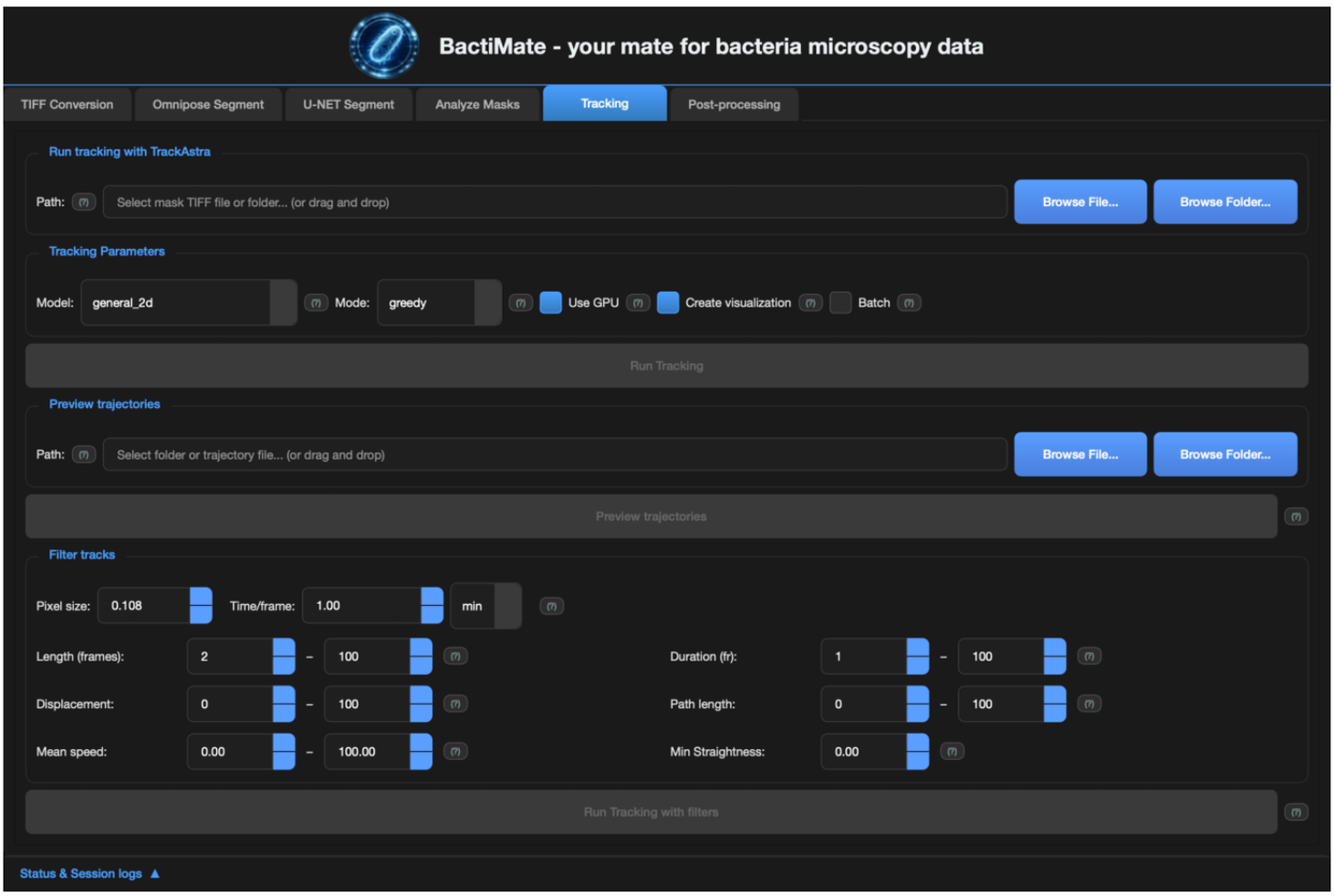


Filter by morphology and size, export to Excel. Optional FI and foci detection.

- Morphological filters: aspect ratio, eccentricity, solidity; size filters (area, width, length)
- Export to Excel: Summary + Per-Cell Data; filter visualization TIFF
- Analyze FI (fluorescence intensity); match by name or one file for all
- Run foci analysis: Manual, Auto-detect, or Spotiflow; combined summary across folder

Tracking

Follow every cell. Every frame.



Trackastra delivers cell lineage and persistent IDs. Link to images and Excel.

- Mask TIFF (time series) required; optional original TIFF and Excel from Analyze Masks
- Model: general_2d or general_2d_w_SAM2_features; Mode: greedy, greedy_nodiv, or ilp
- Run Tracking → tracked masks, lineage, persistent cell IDs
- Device: CPU, CUDA, or Apple MPS

Post-processing

Publication-ready in minutes.



Scale bars, montages, growth-curve videos. From data to figure in one place.

- Add scale bar: length, position, color; pixel size from metadata
- Montage: folder or files → grid (phase, FI channels, merged); labels, spacing
- Growth curve video: TIFF sequence + Excel → side-by-side graph and video
- Output: TIFF, AVI, GIF, MP4; time per frame and FPS settings